PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

ARUSHA REGION

FORM TWO PRE – MOCK EXAMINATION

043 **MATHEMATICS**

Time: 2:30 Hours MARCH 2026

INSTRUCTIONS

1. This paper consists of ten (10) questions
2. Each question carries ten (10) marks
3. Answer All questions and show your work clearly
4. Non – programmable may be used.

NO.	SCORE	SIGNATURE
1.		
2.		
3.		
4.		
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6.		
7.		
8.		
9.		
10.		
TOTAL		

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CHECKER'S SIGN	

1. (a) Explain with vivid examples how each of the following subjects are connected to mathematics. (Give two points each subject)
   (i) Physics
   (ii) Business studies

   (b) A monthly Salary of a certain worker is Tshs. 250,000. One – fifth of his Salary is spent in buying car fuels, half the remaining salary is spent in buying food and half of the rest is spent in miscellaneous expenses

   (i) How much is his monthly savings?

   (ii)State whether the answer in part (i) is a rational number or not.

2. (a) If A: B = 2:3 and B:C= 5:6. Find A: C.
   (b)A special cereals contains rice, wheat and corn in the ratio of 2:3:5 respectively. If a bag of a mixture contains 3 Kilograms of rice. How much corn does it contains?
   (c)Change $0.5\dot{6}$ into the form of $^{a}/_{b}$ in its simplest form, where b≠ 0.

3. (a) Represent the solution of the following inequality on a number line $|2x + 1| \leq 3$
   (b)Round off
      (i) 6.645 to the nearest hundredths
      (ii)8,125,600 to the nearest ten thousand.
   (c)Simplify 4 x -3 (2x -5).
4. (a) Two complementary angles are such that one angle is twice the other. Find the size of those angles.
   (b) Given that $\frac{AB}{KL} = \frac{BC}{LT} = \frac{AC}{KT} = 3$
   (i) Name the similar triangles
   (ii) Name corresponding angles
5. (a) Solve for x in the equations

$$\frac{x-6}{7} - \frac{2x+3}{4} = \frac{x-5}{2}$$

   (b)Solve simultaneously equations

$$\begin{cases} 2x + y = 8 \\ x - y = 1 \end{cases}$$

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6. (a) Given a straight line. $4x + 5y = 40$
   (i) Express this equation in the form of $y = mx + c$
   (ii) What is the gradient of that line
   (iii) Find x and y- intercepts of the line.

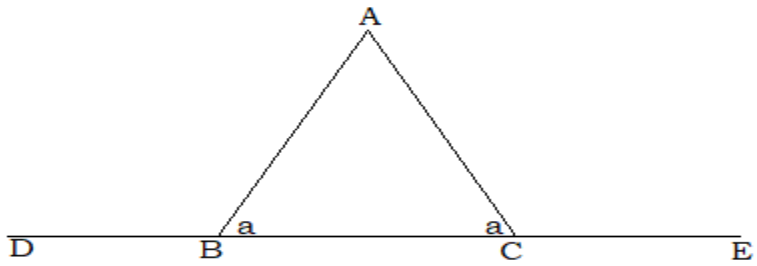
   (b) Determine the value of K in order that the straight line whose equation is

   $kx - y = 5$ passes through the point (3,5)

7. (a) The cost of apples, C varies directly as the mass, m, in Kilograms. I 5kg of apples cost Tsh. 15,000. Find the costant of variation, K.
   (b) The variable y is inversely proportional to the square of x. when $x = 3$, $y = 4$
   (i) Find the equation which connecting x and y
   (ii) Calculate the value of y when $x = 2$
   (c)If 8 workers can complete a certain job in 15 hours
   (i) Write the mathematical relationship between workers and hours
   (ii)How many hours will it take 12 workers to complete the same job?

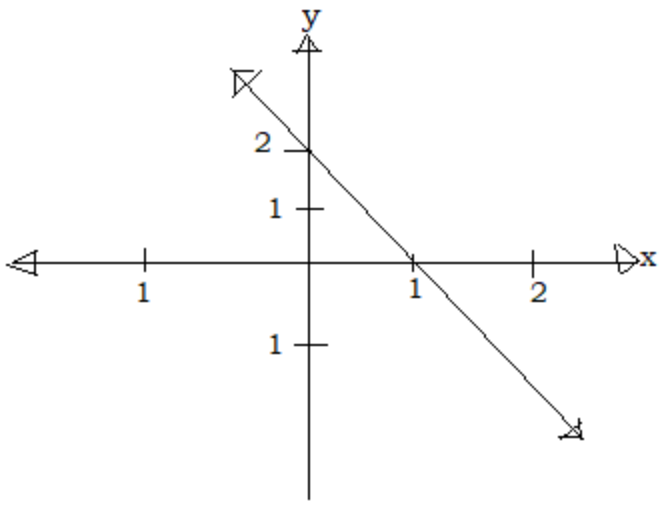
8. (a) Triangle ABC is an isosceles triangle in which AB and AC are equal. If D is the midpoint of AB. Prove that
   (i) $\Delta ABD \equiv \Delta ACD$
   (ii) $A\hat{B}D = A\hat{C}D$

   (b)Use the following figure to prove that $A\hat{B}D = A\hat{C}E$



9. (a) Find the equation of the line in the following graph.

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(b)Find graphically the solution of the following system of linear simultaneous equations.

$$\begin{cases} \frac{x}{2} + \frac{y}{2} = 1 \\ x = 1 + y \end{cases}$$

10. A trader in Dar es salaam bought 50 bags of rice at Tsh 80,000 per bag. He spent Tsh 200,000 on transport. He later sold each bag at Tsh 95,000
    (a) Find total cost price
    (b) Find the total selling price
    (c) Calculate the profit made
    (d) Find the percentage profit

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